

Chapter 1: Introduction to Basic Electronics

1. Definitions

Electronics is the branch of science and technology that deals with the study and application of devices, circuits, and systems that use electrical energy to control, process, and transmit information. Electronics typically involves the flow of electric current through semiconductor materials, which allows for the manipulation of signals in various forms, such as digital, analog, or electromagnetic signals.

An electrical circuit or network is a collection of conductors connected together, generally containing generators, receivers, and resistances.

A dipole is an electrical element capable or not of supplying energy, communicating with the exterior only through two terminals. The resistance or voltage source between points B and M in the figure constitutes two examples of dipoles.

A node is a point in the network where more than two conductors are connected.

A branch is a portion of the network located between two nodes.

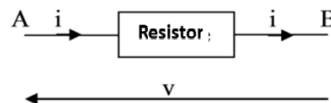
2. Classification of Dipole: We generally categorize dipoles into two main groups:

2.1. Passive Dipoles

These components cannot provide energy to the circuit for an indefinite period. They either dissipate energy or store it in a field.

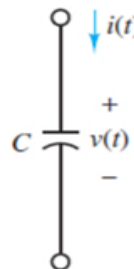
- **Resistors:** Dissipate energy as heat (Joule effect).

$$V=R.I$$



- **Capacitors (Condensateurs):** Store energy in an electric field.

$$i(t) = C \cdot \frac{dv(t)}{dt}$$

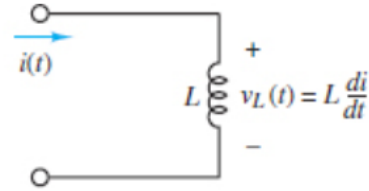


C: Capacitance (measured in **Farads**, F)

- **Inductors (Inductances / Bobines):** Store energy in a magnetic field.

$$v(t) = L \cdot \frac{di(t)}{dt}$$

- L : Inductance (measured in **Henrys**, H).



2.2. Active Dipoles: These components can supply electrical energy to the circuit.

- **Voltage Sources:** Maintain a specific voltage (e.g., a battery).
- **Current Sources:** Maintain a specific flow of current.



The symbolic representation of the ideal and practical : a) voltage source b) current source

3. Classification of Circuit Components: Linear and Non-linear

- **BehaviorLinear:** The relationship between V and I is a straight line ($V = K.I$).
- **Non-linear:** The relationship is more complex (**Diode**, which only allows current to flow in one direction).

Category	Examples	Role in Circuit
Passive	Resistor, Capacitor, Inductor	Consumes or stores energy
Active	Battery, Solar cell, Dynamo	Supplies energy
Non-linear	Diode, Zener diode	Controls/rectifies current flow

4. Kirchhoff's laws

- **Node Law**

The algebraic sum of all currents at a node is zero. In other words, the sum of currents entering a node is equal to the sum of currents leaving that node.

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- **Loop (Mesh) Law**

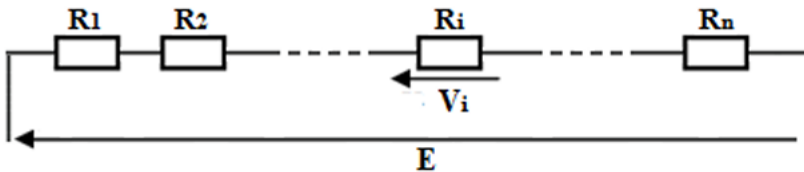
The algebraic sum of the voltages around any mesh, counted in a given direction, is zero at all times.

Application: Exercise 1

5. Fundamental Theorems

5.1 The Voltage Divider Theorem

Let n resistances be connected in series and powered by a voltage E



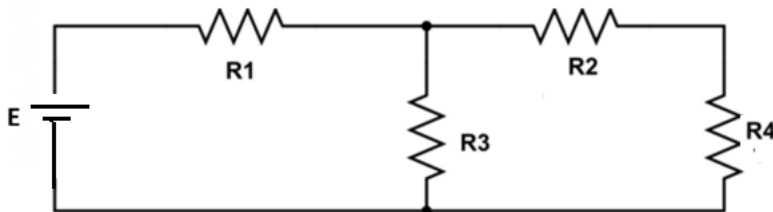
The voltage across the i -th resistance R_i is given by:

$$V_i = \frac{R_i}{\sum_{j=1}^n R_j} E$$

Example:

-Calculate V_2 et V_4 using divider tension theorem.

Give: $V_1 = 20V$, $R_1 = 10 \Omega$, $R_2 = 5 \Omega$, $R_3 = 30 \Omega$ and $R_4 = 10 \Omega$



Attention

~~$$V_{R3} = \frac{R_1}{R_1 + R_3} V_1$$~~

Resistors R_2 and R_4 are in series; therefore, their equivalent resistance R_{24} is:

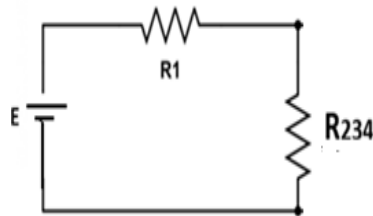
$$R_{24} = R_2 + R_4 = 5\Omega + 10\Omega = 15\Omega$$

The resistors R_2 and R_4 are in series, so their equivalent resistance R_{24} is:

$$R_{24} = R_2 + R_4 = 5\Omega + 10\Omega = 15\Omega$$

This R_{24} is in parallel with R_3 , so their combined resistance R_{234} is:

$$R_{234} = \frac{R_{24} \times R_3}{R_{24} + R_3} = \frac{15\Omega \times 30\Omega}{15\Omega + 30\Omega} = \frac{450\Omega^2}{45\Omega} = 10\Omega$$



Using the voltage divider rule within the series combination of R_2 and R_4 (using $V_{parallel} = E$ as the source for this branch):

Voltage across R_2 :

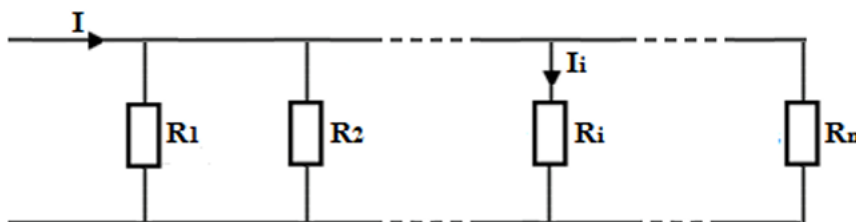
$$V_2 = E \times \frac{R_2}{R_2 + R_4} = 10V \times \frac{5\Omega}{5\Omega + 10\Omega} = 10V \times \frac{5\Omega}{15\Omega} \approx 3.33V$$

Voltage across R_4 :

$$V_4 = E \times \frac{R_4}{R_2 + R_4} = 10V \times \frac{10\Omega}{5\Omega + 10\Omega} = 10V \times \frac{10\Omega}{15\Omega} \approx 6.67V$$

5.2 The Current Divider Theorem

Let n resistances be connected in parallel and powered by a current I



The current I_i passing through the i -th resistance R_i is given by:

$$I_i = \frac{\frac{1}{R_i}}{\sum_{j=1}^n \frac{1}{R_j}} I$$

Application: Exercise 3

5.3 Superposition Theorem

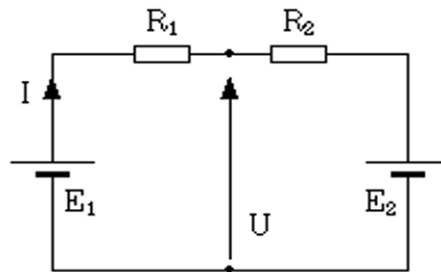
Definition

This theorem is fundamental. It allows us to study circuits with multiple generators (voltage or current sources) by considering the influence of each generator independently from the others, which greatly simplifies most problems.

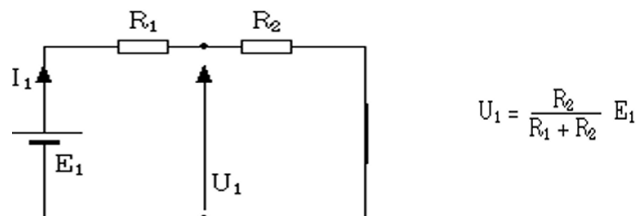
Theorem:

In a circuit with multiple generators, the solution to the problem (the unknown voltages and currents) is the sum of the solutions found by considering one generator at a time. To do this, replace each ideal voltage source with a short circuit and each current source with an open circuit, except for the source whose influence we want to determine.

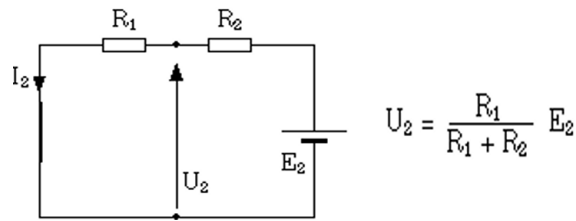
Example



In the example above, we will start by removing E_2 and calculate the voltage U with E_1 alone. We then have a voltage divider:



To find the contribution of E_2 , we then do the same thing by removing E_1 :



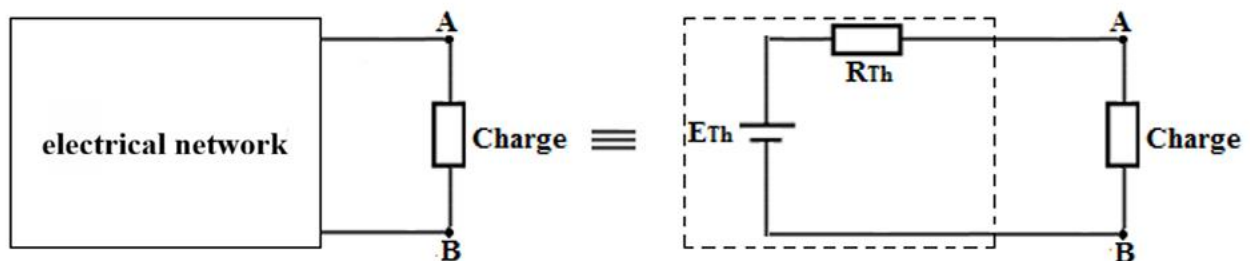
The total solution U is equal to the sum of the two solutions previously found:

$$U = \frac{R_2}{R_1 + R_2} E_1 + \frac{R_1}{R_1 + R_2} E_2$$

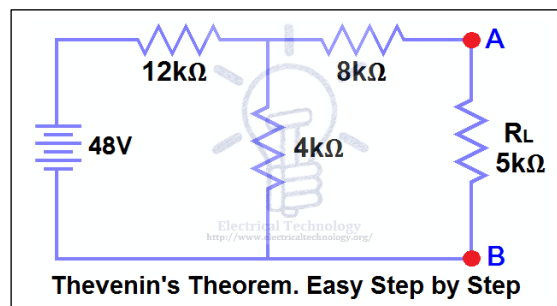
5.4 Thevenin's Theorem

A linear electrical circuit placed between two points A and B can be replaced with a Thévenin equivalent generator with an electromotive force (EMF) E_{TH} and an internal resistance R_{TH} with respect to these points.

- E_{TH} is equal to the open-circuit voltage between points A and B (the voltage V_{AB} when the load is disconnected).
- R_{TH} is the equivalent resistance seen between points A and B when the load is disconnected and the sources are deactivated

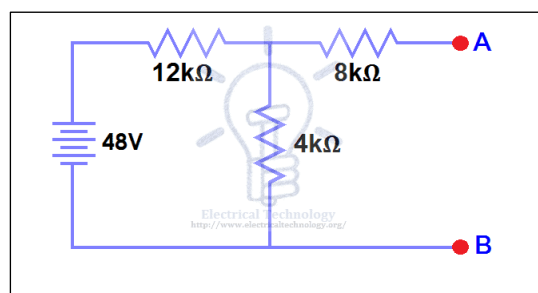


Example



Step 1.

Open the 5kΩ load resistor : **Step 2.**



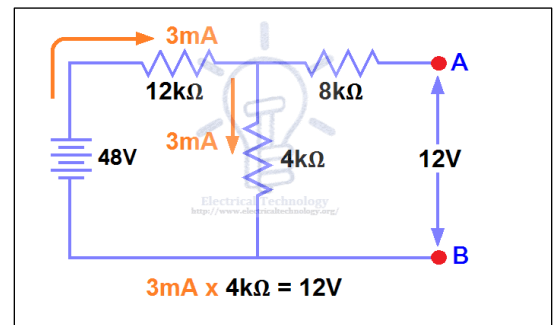
- **Open Circuit Condition:** The load resistor is removed, creating an open circuit between terminals A and B.
- **Current Analysis:** A current of **3mA** flows through the **12 kΩ** and **4 kΩ** resistors because they are in series.
- **The 8 kΩ Branch:** No current flows through the **8 kΩ** resistor since it is part of an open branch; however, it remains connected to the node shared with the **4 kΩ** resistor.
- **Voltage Calculation:**

The voltage drop across the $4\text{ k}\Omega$ resistor is calculated using Ohm's Law:

$$V = I \times R = 3\text{mA} \times 4\text{k}\Omega = 12\text{V}$$

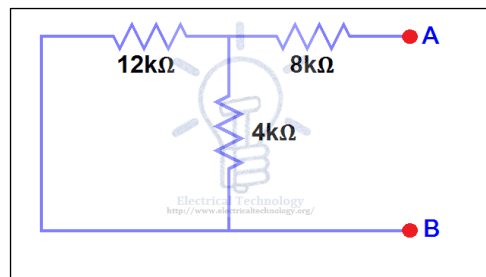
Since there is no voltage drop across the **8 kΩ** resistor (because $I=0$), the voltage at terminals A and B is identical to the voltage across the **4 kΩ** resistor.

- $V_{TH} = 12\text{V}$



Step 3.

Open all current sources and short all voltage sources, as shown in Figure:



Step 4.

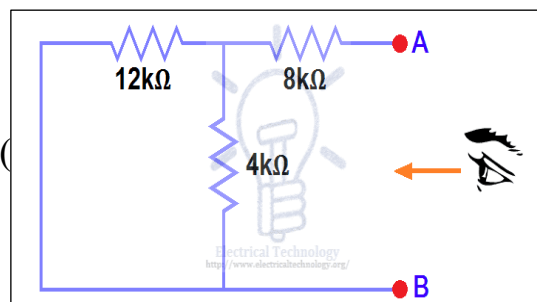
Calculate the open circuit resistance. This is the Thevenin Resistance (R_{TH}).

We have removed the 48V DC source to zero by replacing it with a short circuit in Step 3.

We can see that the **8kΩ** resistor is in series with the parallel combination of the **4kΩ** resistor and the **12kΩ** resistor:

$$R_{TH} = 8\text{k}\Omega + (4\text{k}\Omega \parallel 12\text{k}\Omega)$$

$$R_{TH} = 8\text{k}\Omega + [(4\text{k}\Omega \times 12\text{k}\Omega) / (4\text{k}\Omega + 12\text{k}\Omega)]$$

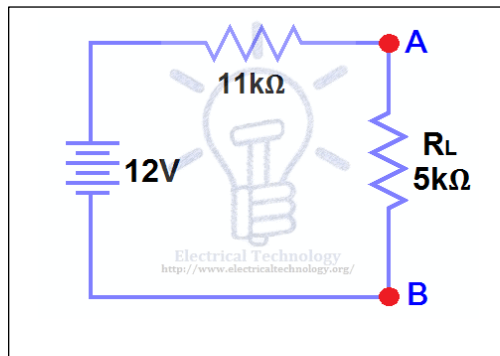


$$R_{TH} = 8k\Omega + 3k\Omega$$

$$R_{TH} = 11k\Omega$$

Step 5

Connect the R_{TH} in series with voltage source V_{TH} and re-connect the load resistor. This is shown in figure. Thevenin circuit with load resistor. This is the Thevenin's equivalent circuit.



Step 6.

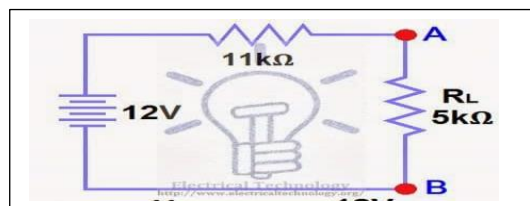
Now, apply the last step i.e. [Ohm's law](#) . Calculate the total load current and load voltage as shown in figure :

$$I_L = V_{TH} / (R_{TH} + R_L) \quad I_L = 12V / (11k\Omega + 5k\Omega) \rightarrow = 12/16k\Omega$$

$$I_L = 0.75mA \quad \text{And} \quad V_L = I_L \times R_L$$

$$V_L = 0.75mA \times 5k\Omega$$

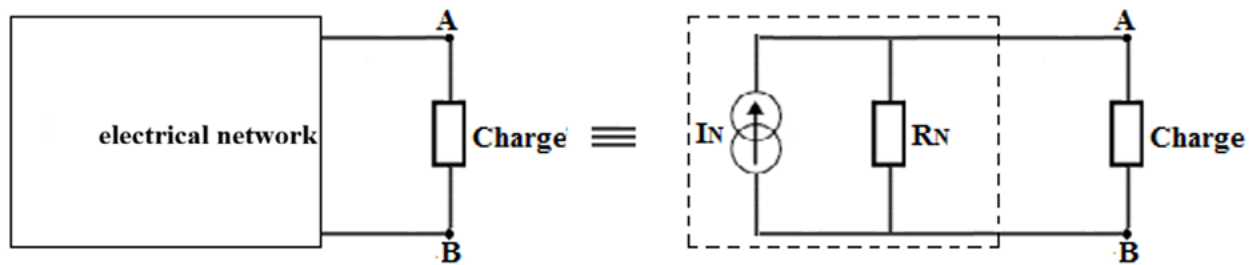
$$V_L = 3.75V$$



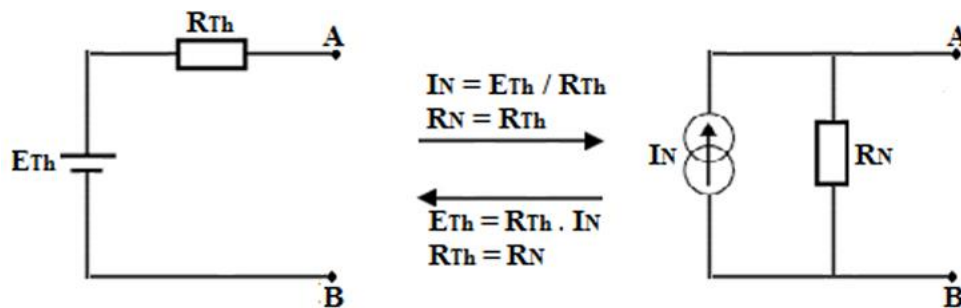
5.5 Norton Theorem

The **Norton Theorem** states that a linear electrical circuit placed between two points A and B can be replaced by an equivalent Norton generator with a current source I_N and an internal resistance R_N with respect to these points.

- I_N is equal to the short-circuit current between points A and B (the current when the load is short-circuited).
- R_N is determined in the same way as the Thévenin resistance R_{TH} .



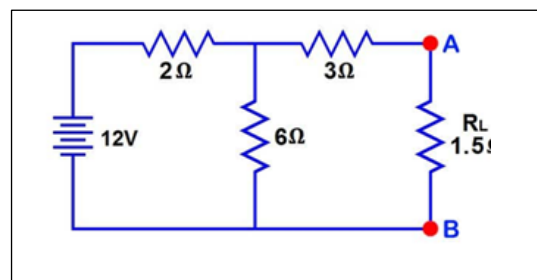
Equivalence between Thévenin and Norton



Example

Using Northern theorem :

- Calculate current flowing through the load resistor I_L :
- Calculate voltage across the load resistor

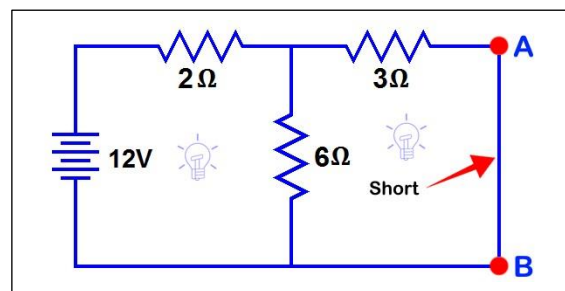


Solution:

Step1.

Short the 1.5Ω load resistor as shown in Figure .

Step 2.



Measure the Short Circuit Current. This is the Norton Current (I_N).

To determine the Norton current (I_N), we short the AB terminals. The 6Ω and 3Ω resistors are now in parallel, and this parallel combination is in series with the 2Ω resistor. So, the total resistance of the circuit to the source is:

$$R_T = 2\Omega + (6\Omega \parallel 3\Omega)$$

$$R_T = 2\Omega + [(3\Omega \times 6\Omega) / (3\Omega + 6\Omega)] \rightarrow I_T = 2\Omega + 2\Omega = 4\Omega.$$

$$R_T = 4\Omega \quad I_T = V \div R_T \quad I_T = 12V \div 4\Omega$$

$$I_T = 3A.$$

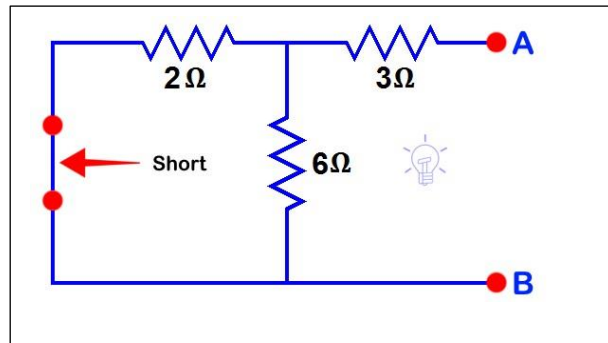
Now we have to find $I_{SC} = I_N$ (Current Divider Rule):

$$I_{SC} = I_N = 3A \times [(6\Omega \div (3\Omega + 6\Omega))] = 2A.$$

$$I_{SC} = I_N = 2A.$$

Step 3

Open the current source, short the voltage source, and remove (open) the load resistor, as shown in Figure.



Step 4

Calculate /measure the open circuit resistance. This is the Norton Resistance (R_N)

In Step (3), we reduced the 12 V DC source to zero, which is equivalent to replacing it with a short circuit, as shown in Figure. We can see that the 3Ω resistor is in series with the parallel combination of the 6Ω and 2Ω resistors :

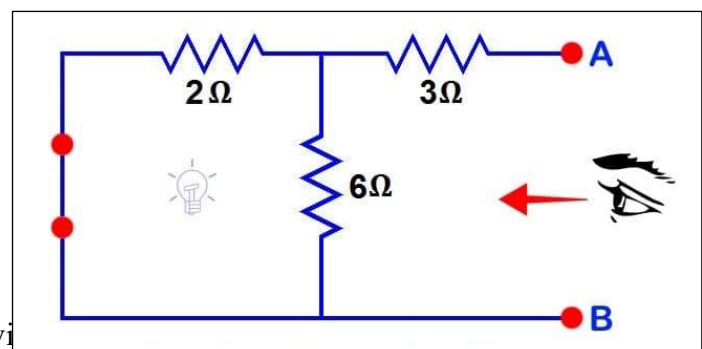
$$R_N = 3\Omega + (6\Omega \parallel 2\Omega)$$

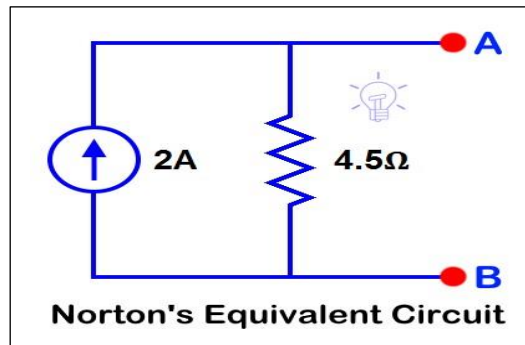
$$R_N = 3\Omega + [(6\Omega \times 2\Omega) \div (6\Omega + 2\Omega)]$$

$$R_N = 3\Omega + 1.5\Omega$$

Step 5: Final Norton Equivalent Circuit

Connect the Norton resistance (R_N) in parallel with the load resistor (R), as shown in Figure 6. This represents the **Norton equivalent circuit** with the load resistor connected.





Step 6: Final Calculations

Finally, apply the last step: calculate the **load current** (I_L) flowing through and the **load voltage** (V_L or U_{AB}) across the load resistor using **Ohm's Law**.

Current flowing through the load resistor I_L :

$$I_L = I_N \times [R_N \div (R_N + R_L)] = 2A \times (4,5\Omega \div 4,5\Omega + 1,5\Omega)$$

$$I_L = 1,5 A$$

Voltage across the load resistor : $V_L = I_L \times R_L$

$$V_L = 1,5 A \times 1,5 \Omega$$

$$V_L = 2,25 V$$

